

Print Function

```
In [4]: 2+3  
        10-6  
        10 * 5
```

Out[4]: 50

```
In [5]: print(2+3)  
        print(10-6)  
        print(10*5)
```

5
4
50

```
In [6]: # print is used to get answer  
        a = 10  
        b = 20  
        print(a)  
        print(b)
```

10
20

```
In [4]: print(a,b)
```

10 20

```
In [2]: a = 10  
        b = 20  
        c = 30  
        print(a)  
        print(b)  
        print(c)  
  
        print(a,b,c) # Code optimization & Memory optimization
```

10
20
30
10 20 30

```
In [6]: print(10)  
        print(10,20)  
        print('Python')  
        print(10,20,'Python')
```

10
10 20
Python
10 20 Python

```
In [1]: num1 = 20  
        num2 = 30  
        add = num1 + num2  
        print(add)
```

50

Print Result with String

```
In [4]: num1 = 20
num2 = 30
add = num1 + num2
print('The addition of', num1, 'and', num2, 'is =', add)
```

The addition of 20 and 30 is = 50

```
In [5]: num1 = 20
num2 = 30
num3 = 50
add = num1 + num2 + num3
print('The addition of', num1, num2, 'and', num3, 'is', add)
```

The addition of 20 30 and 50 is 100

```
In [6]: name = 'Python'
age = 20
city = 'Hyderabad city'
print('My name is', name, 'and I am', age, 'years old, from', city)
```

My name is Python and I am 20 years old, from Hyderabad city

Print() with format method

```
In [2]: num1 = 20
num2 = 30
add = num1 + num2
print('The addition of {} and {} is = {}'.format(num1,num2,add))
```

The addition of 20 and 30 is = 50

```
In [6]: num1 = 20
num2 = 30
num3 = 40
num4 = 50
add = num1 + num2 + num3 + num4
print('The addition of {} and {} and {} and {} is= {}'.format(num1,num2,num3,num4,add))
```

The addition of 20 and 30 and 40 and 50 is= 140

```
In [3]: name = 'Python'
age = 20
city = 'Hyderabad City'
print('Hello my name is {}, and i am {} years old from {}'.format(name, age, city))
```

Hello my name is Python, and i am 20 years old from Hyderabad City

```
In [2]: num1 = 100
num2 = 25
num3 = 333
avg = (num1 + num2 + num3)/3 # The average of num1, num2, num3 is avg.
avg1 = round((num1 + num2 + num3)/3,2) # Can use it to get average also

print('The average of {}, {} and {} is = {} or {}'.format(num1, num2, num3, avg, avg1))
```

The average of 100, 25 and 333 is = 152.66666666666666 or 152.67

```
In [3]: round(avg, 2)    # Round of till 2 digit after decimal
```

```
Out[3]: 152.67
```

More short format method (f string method)

```
In [6]: num1 = 20
num2 = 30
add = num1 + num2
print(f'The addition of {num1} and {num2} is = {add}')
```

The addition of 20 and 30 is = 50

```
In [10]: name = 'Python'
age = 20
city = 'Hyderabad city'
print(f'My name is {name}, and i am {age} years old, from {city}.')
```

My name is Python, and i am 20 years old, from Hyderabad city.

```
In [11]: num1 = 100
num2 = 25
num3 = 333
avg = (num1 + num2 + num3)/3
avg1 = round((num1 + num2 + num3)/3,2)
print(f'The average of {num1}, {num2} and {num3} is {avg} or {avg1}.')
```

The average of 100, 25 and 333 is 152.66666666666666 or 152.67.

```
In [12]: # Let's combine all
```

```
In [14]: num1 = 10
num2 = 20
add = num1 + num2
print('The additon of', num1, 'and', num2,'is', add)
print('The addition of {} and {} is {}'.format(num1, num2, add))
print(f'The addition of {num1} and {num2} is {add}.')
```

The additon of 10 and 20 is 30
 The addition of 10 and 20 is 30
 The addition of 10 and 20 is 30.

End Statement

```
In [7]: print('Hello')    #1st statement
print('Good evening')    #2nd statement

# I want to print Like:- Hello Good evening
```

Hello
 Good evening

```
In [15]: print('Hello', end=' ')    #1st statement
print('World Good day')    #2nd statement
```

Hello World Good day

```
In [16]: print('Hello', end=' ### ')
         print('World Good day')
```

Hello ### World Good day

Separator

```
In [18]: print('Hello','Hey','How are you', sep = '--->')
```

Hello--->Hey--->How are you

```
In [19]: print('Hello','Hey','How are you', sep = '*****')
```

Hello*****Hey*****How are you

```
In [20]: print('Hello','Hey','How are you', sep = '&')
```

Hello&Hey&How are you

```
In [17]: print('Hello','Hey','How are you', sep = '@')
```

Hello@Hey@How are you

```
In [21]: print('Hello','Hey','How are you', sep = ' ')
```

Hello Hey How are you

```
In [8]: print(3, '.')    # . is far from 3 so here we'll use sep method
```

3 .

```
In [9]: print(3, '.', sep='')    # See now space setteld (also use to remove space between
```

3.

```
In [10]: print(1,2,end = ' ')
         print(3, '.', sep = ' ')
```

Will print 1 2 3 .

1 2 3 .

Operators

1. Arithmetic Operators

```
In [14]: x1, y1 = 10, 5
```

```
In [15]: x1 + y1    # Addition
```

Out[15]: 15

```
In [16]: x1 - y1    # Substraction
```

Out[16]: 5

```
In [17]: x1 * y1    # Multiplication
```

Out[17]: 50

```
In [18]: x1 / y1    # Float division
```

Out[18]: 2.0

```
In [19]: x1 // y1  # Int division
```

Out[19]: 2

```
In [20]: x1 % y1   # Module
```

Out[20]: 0

```
In [21]: x1 ** y1  #power
```

Out[21]: 100000

```
In [22]: 3 ** 2    # 3 to the power of 2
```

Out[22]: 9

```
In [23]: x2 = 3
y2 = 3

x2 ** y2
```

Out[23]: 27

2. Assignment Operator

```
In [13]: x = 2
```

```
In [33]: x = x + 2    # If you want to increment by 2
x
```

Out[33]: 4.0

```
In [34]: x += 2    # Means x = x + 2
x
```

Out[34]: 6.0

```
In [35]: x -= 2    # x = x - 2
x
```

Out[35]: 4.0

```
In [36]: x *= 2    # x= x * 2
x
```

Out[36]: 8.0

```
In [37]: x /= 2    # x = x / 2 (float division)
```

```
x
```

Out[37]: 4.0

```
In [38]: x **= 2    # x = x to the power 2
x
```

Out[38]: 16.0

```
In [39]: x //= 2    # x = x // 2 (int division)
x
```

Out[39]: 8.0

```
In [40]: a, b = 5, 6    # You can assigned variable in one line as well
print(a)
print(b)
```

5
6

```
In [41]: a = 5
b = 6
print(a)
print(b)
```

5
6

3. Unary Operator

```
In [42]: n = 7    # Negation
n
```

Out[42]: 7

```
In [43]: m = -(n)
m
```

Out[43]: -7

```
In [44]: n
```

Out[44]: 7

```
In [45]: -n
```

Out[45]: -7

4. Relational Operator

```
In [46]: a = 5
b = 6
a < b
```

Out[46]: True

In [47]: `a > b`

Out[47]: False

In [48]: `print(a)`
`print(b)`

5

6

In [49]: *# a = b # We can't use '=' operator that means it is assigning*

In [50]: `a == b`

Out[50]: False

In [51]: `a != b`

Out[51]: True

In [54]: *# Here, if i change b = 5*
b = 5

In [55]: `a == b`

Out[55]: True

In [56]: `a`

Out[56]: 5

In [57]: `b`

Out[57]: 5

In [58]: `a != b`

Out[58]: False

In [59]: `a1, b1 = 6, 7`

In [60]: `a1 > b1`

Out[60]: False

In [62]: `a1 >= b1`

Out[62]: False

In [63]: `a1 < b1`

Out[63]: True

In [64]: `a1 <= b1`

Out[64]: True

In [66]: `a1 == b1`

Out[66]: False

In [67]: `a1 != b1`

Out[67]: True

5. Logical Operator

(i) AND Operator

In [68]: `a = 5`
`b = 4`

In [69]: `a < 8 and b < 5` *#refer to the truth table*

Out[69]: True

In [70]: `a < 8 and b > 5`

Out[70]: False

(ii) OR Operator

In [71]: `a < 8 or b < 2`

Out[71]: True

In [72]: `a > 8 or b < 2`

Out[72]: False

(iii) NOT Operator

In [73]: `x = False`
`x`

Out[73]: False

In [74]: `not x` *# You can reverse the operation*

Out[74]: True

In [75]: `x = not x`
`x`

Out[75]: True

```
In [76]: y = True  
y
```

```
Out[76]: True
```

```
In [77]: not y
```

```
Out[77]: False
```

```
In [ ]:
```